

MARK DINOL RODRIGO

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SUMMARY

Driven Computer Science undergraduate specializing in backend system design, algorithmic problem-solving, and machine learning. Proven track record in competitive programming, ranking in the Top 10% globally on LeetCode. Competent in translating machine learning research paper concepts into functional TensorFlow implementations, alongside building full-stack applications using Python, Java, and the MERN stack.

EDUCATION

University of Westminster

Sep 2024 - Present

BSc Computer Science — Undergraduate

Key Modules: *Object-Oriented Programming (94), Python Programming (96), Web Application Development (85), Database Systems (93), Mathematics for computing (97), Data Structures & Algorithms, Machine Learning, Client-Server Computing*

Informatics Institute of Technology

Sep 2023- Present

Foundation Certificate in Higher Education (Distinction)

St. Peter's College, Negombo

Jan 2012 - Jun 2023

GCE Ordinary Level Examination (9As)

Subjects: All the core subjects and *ICT, Business & Accounting Studies and Art*

SKILLS

Languages : Python, Java, JavaScript, Rust

Frameworks & Runtimes : React, Node.js, Express, Flask, JAX-RS, Tauri

Databases, Cloud & AI : TensorFlow, MongoDB, Firebase, Google Cloud, Elasticsearch

Core Concepts : LLM Fine-Tuning (LoRA), Knowledge Distillation, Machine Learning, Database Systems, OOP

Tools & Design: Git, Figma

PROJECTS

Mom OS | Python, Tauri, ElasticSearch, Rust

- Built a behavioral analysis system to track and evaluate user activity patterns in real time.
- Designed and implemented REST APIs to process and manage user interaction data.
- Optimized Elasticsearch queries to improve data retrieval performance for analytics.
- Developed a desktop interface using Tauri, integrating backend logic with a responsive frontend.

Vector Search Engine (HNSW) from Scratch | Python

- Engineered a Hierarchical Navigable Small World (HNSW) graph-based index from scratch for high-dimensional approximate nearest neighbor (ANN) search.
- Implemented multi-layer graph traversal, heuristic-based neighbor selection, and a priority-queue-backed search mechanism.

Notewave | React, Node.js, Firebase

- Built a full-stack application combining messaging and video call features to replace fragmented solutions like WhatsApp + Zoom in online education.
- Led a 5 member team and managed to get included in the **semi-finals of Codesprintx Sri Lanka**.

ACHIEVEMENTS

- Semi-finalist in Codesprintx Sri Lanka for the Notewave project. (led a 5-member team)
- Ranked in the top 10% globally on LeetCode for competitive programming.
- Completed Deep Learning and Machine Learning specializations by Andrew Ng.
- Obtained Hackeerrank's problem solving certifications. (Beginner, Intermediate)

VOLUNTEERING / EXTRACURRICULAR

Environmental Prefect – St. Peter's College, Negombo | 4 years

- Led and organized school-wide environmental initiatives, promoting sustainability and awareness among students.
- Awards: Earned 3 badges from the Central Environmental Authority of Sri Lanka for completing the required activities for each badge.

INTERESTS

Chess (1900 rapid in chess.com), **Competitive Programming** (LC 1700+), **Painting**

REFEREES

Indrajith Ekanayake

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